

IMAT MOCK EXAM 7 — "The Real Thing"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 38/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Logical Reasoning & Reading (Q1-9)

1. "Resistance to antibiotics commonly arises when random mutations reduce a bacterium's sensitivity to a drug. Under antibiotic treatment, resistant cells survive while their susceptible competitors are killed, so resistant strains spread rapidly. Resistance genes can also move between bacterial species on plasmids. Long-term studies of sites where antibiotic use was halted years ago show that resistant strains sometimes remain common for a very long time, apparently because the fitness cost of resistance was offset by compensatory mutations."

Which statement is most strongly supported by the passage?

- A)** Antibiotics deliberately induce resistance mutations in bacteria.
- B)** Stopping antibiotic use does not guarantee that resistance will disappear.
- C)** Resistant bacteria are always fitter than susceptible ones.
- D)** Resistance genes cannot transfer between species.
- E)** Plasmids are a kind of virus.

Answer: B

The passage's final sentence reports that years after antibiotic use stopped, resistant strains were still common thanks to compensatory mutations, so halting use does not guarantee resistance dies out. The claim that antibiotics deliberately induce mutations contradicts the passage's statement that resistance mutations arise randomly; the claim that resistant bacteria are always fitter is contradicted by the fitness cost that had to be offset; calling plasmids a kind of virus is unsupported; and denying transfer between species contradicts the plasmid sentence directly.

2. A randomised trial reports that patients given drug X recovered in a median of 5 days versus 7 days with placebo, recovery being recorded by patient self-report. The authors conclude X speeds recovery.

Which of the following, if true, most weakens the authors' conclusion?

- A)** A few patients in each arm dropped out before day 7.
- B)** The placebo tablets were white and the X tablets were also white.
- C)** The trial lasted six months in total.
- D)** The trial ran in two countries at once.
- E)** Drug X is intensely bitter and clearly distinguishable from the tasteless placebo, so patients knew which arm they were in.

Answer: E

Recovery is recorded by patient self-report, so the result is trustworthy only if patients cannot tell which arm they are in. An intensely bitter drug next to a tasteless placebo unblinds every participant, so expectation effects can produce or exaggerate self-reported faster recovery, an alternative explanation needing no pharmacological action at all. A few symmetric dropouts, two countries, identical white tablets and the total trial length either affect both arms equally or preserve the blind.

3. (General knowledge) The 2025 G7 summit was hosted by:

- A) Japan
- B) Italy
- C) France
- D) Canada
- E) the United Kingdom

Answer: D

The G7 presidency rotates yearly: 2023 Japan, 2024 Italy (Apulia), 2025 Canada, which hosted the summit at Kananaskis, Alberta. The decisive fact is simply the 2025 host: Canada.

4. A clinic must schedule three locums, X, Y and Z, one for each of Monday, Tuesday and Wednesday. X must work earlier in the week than Z. Y cannot work on Tuesday.

Which of the following must be true?

- A) X does not work on Wednesday.
- B) X works on Monday.
- C) Z works on Wednesday.
- D) Y works on Monday.
- E) Y works on Wednesday.

Answer: A

Step 1: X must come before Z, so X cannot be on Wednesday (no later day remains for Z). Step 2: if X is Monday, Y cannot take Tuesday, so Z takes Tuesday and Y Wednesday; if X is Tuesday, Z must take Wednesday and Y takes Monday. Step 3: exactly two schedules exist, (X Mon, Z Tue, Y Wed) and (Y Mon, X Tue, Z Wed). In both, X sits on Monday or Tuesday, so X is never on Wednesday; every other listed claim fails in one of the two schedules (for instance Y is on Wednesday in the first schedule but on Monday in the second).

5. A chemical test turns blue if and only if protein X is present in a sample. A particular sample did not turn the test blue.

Which conclusion is valid?

- A) The sample contains protein X.
- B) The sample contains some other protein in quantity.
- C) The sample does not contain protein X.
- D) The test must have been performed incorrectly.
- E) Protein X turns the test red.

Answer: C

'Blue if and only if protein X is present' makes blue both necessary and sufficient for X. The sample did not turn blue, so by the contrapositive of 'X present implies blue', the sample does not contain X. Claims about a botched test, a red colour or other proteins in quantity add assumptions the statement does not license.

6. A clock reads 9:18. What is the angle between the hour hand and the minute hand?

- A) 144° B) 162° C) 171° D) 174° E) 180°

Answer: C

Minute hand: $18 \text{ min} \times 6^\circ/\text{min} = 108^\circ$ from 12.

Hour hand: $9 \times 30^\circ + 18 \times 0.5^\circ = 270^\circ + 9^\circ = 279^\circ$

Angle = $279^\circ - 108^\circ = 171^\circ$

The value 162° comes from freezing the hour hand at 9:00 ($270^\circ - 108^\circ$) and forgetting it advances 0.5° per minute.

7. Of 1,000 people screened, 100 have the disease. The test is positive in 90 of the 100 diseased people, and also positive in 180 of the 900 healthy people.

What fraction of all positive results are false positives?

- A) $1/3$ B) $1/5$ C) $1/2$ D) $2/3$ E) $9/10$

Answer: D

True positives: 90; false positives: 180; all positives: $90 + 180 = 270$.

False share = $180 \div 270 = 2/3$

Because only 10 percent of those screened are diseased, positives from healthy people outnumber true positives two to one. The fraction $180/900$ ($= 1/5$) is the false-positive rate among the healthy, a different quantity with a different denominator.

8. A cyclist rides from A to B at 60 km/h and returns immediately along the same route at 120 km/h. The average speed for the whole trip is:

- A) 60 km/h B) 75 km/h C) 80 km/h D) 90 km/h E) 100 km/h

Answer: C

Average speed = total distance $\div$ total time. Take d km each way: time = $d/60 + d/120 = 3d/120 = d/40$ h; total distance = $2d$.

Average = $2d \div (d/40) = 80$ km/h, the harmonic mean $2 \times 60 \times 120 \div 180$

The simple midpoint 90 km/h ignores that far less time is spent at 120 km/h than at 60.

9. A sequence begins 2, 3, 5, 9, 17, 33, ... Each term follows one rule from the previous term. What is the next term?

- A) 49 B) 63 C) 64 D) 65 E) 66

Answer: D

Successive gaps are 1, 2, 4, 8, 16: each doubles, so the next gap is 32 and the next term is $33 + 32 = 65$. Equivalently, every term is twice the previous one minus 1: $2 \times 33 - 1 = 65$.

Section B - Biology (Q10-32)

10. A plant with genotype AaBb for two unlinked genes is crossed with a double-recessive aabb plant, and the offspring phenotypes appear 1:1:1:1. This result shows that the AaBb parent:

- A)** produces only two kinds of gametes
- B)** must have undergone a mutation during meiosis
- C)** has its two genes tightly linked on one chromosome
- D)** is homozygous dominant at both genes
- E)** is heterozygous at both genes and passes the four allele combinations into gametes equally

Answer: E

The aabb parent can pass on only the ab combination, so each offspring phenotype directly reveals one gamete from the AaBb parent. Four phenotypes in equal numbers mean four gamete types (AB, Ab, aB, ab) at 1/4 each: the signature of independent assortment with double heterozygosity. Tight linkage would instead give mostly the two parental classes, and producing only two gamete types would give only two phenotypes.

11. In snapdragons, flower colour shows incomplete dominance: CRCR is red, CRCW is pink, CWCW is white. Two pink plants are crossed. The expected offspring ratio is:

- A)** 1 red : 2 pink : 1 white
- B)** all pink
- C)** 9:3:3:1
- D)** 3 red : 1 white
- E)** 1 red : 1 white

Answer: A

Pink = CRCW, so the cross is CRCW × CRCW: offspring draw CR or CW from each parent, giving CRCR 1/4, CRCW 1/2, CWCW 1/4, i.e. 1 red : 2 pink : 1 white. Incomplete dominance makes the heterozygote visibly intermediate, so the genotypic 1:2:1 shows up directly as the phenotypic ratio. The 3 red : 1 white ratio belongs to complete dominance, where red would mask white.

12. Two genes assort independently. A cross AaBb × aabb is performed. What fraction of the offspring is expected to be aabb?

- A)** 0 **B)** 1/16 **C)** 1/4 **D)** 1/8 **E)** 1/2

Answer: C

The AaBb parent makes gametes AB, Ab, aB, ab at 1/4 each; the aabb parent makes only ab. An aabb offspring requires ab from each side: probability 1/4. The value 1/16 would come from the 4 × 4 gamete grid of two double heterozygotes, which is not this testcross.

13. Two unaffected parents have a child affected by a rare autosomal recessive disease. What is the probability that their next child is affected?

- A)** 0 (immunity has developed)
- B)** 1/2
- C)** 1/4
- D)** 1/8
- E)** 2/3

Answer: C

An affected child must be aa, so each unaffected parent carries one recessive allele: the mating is Aa × Aa. Each pregnancy is independent, and in Aa × Aa the chance of an aa child is 1/4. The 2/3 figure answers a different question (the carrier probability among unaffected children), and previous births do not change the odds.

14. On the lagging strand, DNA polymerase works away from the replication fork, producing short Okazaki fragments. Which enzyme joins these fragments into a continuous strand?

- A)** helicase
- B)** DNA ligase
- C)** single-strand binding protein
- D)** primase
- E)** topoisomerase

Answer: B

Okazaki fragments are separate DNA pieces separated by nicks in the sugar-phosphate backbone. DNA ligase seals those nicks by forming phosphodiester bonds, producing one continuous strand. Helicase unwinds the helix, primase lays RNA primers, topoisomerase relieves supercoiling and single-strand binding proteins hold the strands apart; none of them joins fragments.

15. A DNA template strand contains the triplet 3'-TAC-5'. The mRNA produced from this triplet is:

- A)** 5'-AUG-3' **B)** 5'-ATG-3' **C)** 5'-GUA-3' **D)** 5'-CAT-3' **E)** 5'-UAC-3'

Answer: A

RNA polymerase reads the template 3' to 5' and builds the mRNA antiparallel, 5' to 3', pairing A to the template's T, U to A, G to C. Complementing 3'-T-A-C-5' in the correct antiparallel orientation gives 5'-A-U-G-3', the start codon for methionine. Complementing in the same left-to-right direction instead yields GUA (AUG read backwards); UAC tempts via the anticodon route; and any triplet containing T wrongly keeps DNA's thymine in an RNA strand.

16. Which of the following codons signals termination of translation?

- A)** UGG **B)** UGA **C)** GGG **D)** AUG **E)** UAC

Answer: B

The three stop codons are UAA, UAG and UGA; they code no amino acid and instead bind release factors. Of the codons listed, only UGA is a stop signal. AUG starts translation (methionine), UGG codes tryptophan, UAC tyrosine and GGG glycine.

17. A protein is heated to 70 °C and denatures. What has happened to its structure?

- A)** The protein has become a different protein.
- B)** Individual amino acids have been chemically destroyed.
- C)** The primary structure has been changed.
- D)** Peptide bonds along the backbone have broken.
- E)** Weak interactions (hydrogen bonds, ionic and hydrophobic contacts) have been disrupted while the amino-acid sequence is preserved.

Answer: E

Denaturation disrupts the weak, non-covalent interactions (hydrogen bonds, ionic contacts, hydrophobic interactions) that hold the folded secondary and tertiary structure together, so the chain unfolds. The covalent peptide bonds and the amino-acid sequence are untouched, so the primary structure survives and the protein can sometimes refold on cooling. Claims that peptide bonds broke or amino acids were destroyed describe hydrolysis or degradation, not denaturation.

18. An inhibitor reduces an enzyme's V_{max} by 75% but leaves K_m exactly unchanged. The inhibitor is best described as:

- A)** competitive
- B)** a substrate analogue competing for the active site
- C)** a cofactor
- D)** an allosteric activator
- E)** non-competitive

Answer: E

V_{max} falls to a quarter while K_m stays exactly the same. An unchanged K_m means the enzyme molecules that still work bind substrate with normal affinity, which rules out any inhibitor competing at the active site (competitive inhibition by a substrate analogue raises the apparent K_m and leaves V_{max} reachable at high substrate). Lowered V_{max} with fixed K_m is the signature of a non-competitive inhibitor, which binds elsewhere and knocks out enzyme molecules regardless of substrate concentration.

19. A poison blocks complex I of the mitochondrial electron transport chain. Which statement follows?

- A)** Pyruvate oxidation accelerates to compensate.
- B)** Oxidation of NADH halts, but electrons from $FADH_2$ can still enter at complex II and reach oxygen.
- C)** All electron transport through complexes II, III and IV stops immediately and completely.
- D)** Glycolysis stops.
- E)** ATP synthase reverses and consumes water.

Answer: B

Electrons enter the chain either at complex I (from NADH) or at complex II (from $FADH_2$), and both routes converge downstream on ubiquinone, then complexes III and IV, ending at oxygen. Blocking complex I stops all NADH oxidation but leaves the complex II route open, so $FADH_2$ electrons can still flow to oxygen while fewer protons are pumped. Glycolysis is a cytosolic pathway and does not stop, and complexes II to IV do not shut down when fed from complex II.

20. The Calvin cycle cannot keep running for long in darkness because it directly requires:

- A)** ATP and NADPH supplied by the light reactions
- B)** glucose from starch breakdown
- C)** chlorophyll excitation
- D)** oxygen released by photosystem II
- E)** carbonic acid

Answer: A

The Calvin cycle's reduction step consumes ATP and NADPH that only the light reactions supply, converting glycerate-3-phosphate to the sugar G3P. In darkness the light reactions stop, these pools drain, and the cycle halts. Oxygen is a light-reaction product the cycle does not use, and its carbon entry point is CO_2 , not glucose from starch breakdown.

21. A red blood cell is placed in seawater, which is hypertonic to blood plasma. What happens?

- A)** Nothing: cells are always in osmotic balance.
- B)** The cell divides.
- C)** Sodium ions are pumped out to compensate, so volume is constant.
- D)** Water leaves the cell and it shrinks (crenates).
- E)** Water enters the cell and it swells.

Answer: D

Seawater is hypertonic to the cell: it has the higher solute concentration and the lower water potential, so water moves out of the red cell by osmosis and the cell shrinks and crenates. Water entering and swelling is the response to a hypotonic medium. Mature red cells cannot pump solutes out to restore volume and certainly do not divide.

22. The protein p53 accumulates when DNA is damaged in G1. Its role is to:

- A)** replicate the damaged DNA segment
- B)** halt the cell cycle so repair can occur, triggering apoptosis if damage is unrepairable
- C)** move the cell into S phase regardless of damage
- D)** phosphorylate the nuclear envelope
- E)** push the cell directly into mitosis

Answer: B

p53 accumulates after DNA damage and acts as the guardian of the G1 checkpoint: it halts the cycle (via targets such as p21) so repair systems get time to work, and if the damage is irreparable it triggers apoptosis. Pushing the cell into S phase or mitosis with damaged DNA would propagate mutations, exactly what this tumour suppressor exists to prevent.

23. Nondisjunction occurs during meiosis in a human oocyte, producing an egg with 24 chromosomes. Fertilisation by a normal sperm gives a zygote with:

- A)** 47 chromosomes (a trisomy, such as trisomy 21 if chromosome 21 is involved)
- B)** certain death of the zygote in all cases
- C)** 46 chromosomes, normal
- D)** 45 chromosomes (monosomy)
- E)** 46 chromosomes with a deletion

Answer: A

The nondisjunction egg carries 24 chromosomes (one extra homologue). A normal sperm contributes 23, so the zygote has $24 + 23 = 47$ chromosomes: one chromosome type in three copies, a trisomy. A 45-chromosome (monosomy) zygote would require an egg with only 22, and trisomies are not uniformly lethal, trisomy 21 being a viable example.

24. The rapid rising (depolarisation) phase of a nerve action potential is caused by:

- A)** the sodium-potassium pump running forward at maximum speed
- B)** chloride ions leaving the axon
- C)** the membrane becoming impermeable to all ions
- D)** opening of voltage-gated potassium channels, K^+ entering
- E)** opening of voltage-gated sodium channels, Na^+ rushing in

Answer: E

Threshold depolarisation opens voltage-gated sodium channels, and Na^+ floods into the axon down its electrochemical gradient, driving the membrane potential rapidly positive: that inward sodium current is the rising phase. Potassium exit through voltage-gated K^+ channels causes the falling phase, and the sodium-potassium pump is far too slow to produce a millisecond upstroke.

25. At a chemical synapse, the immediate trigger for neurotransmitter release from the presynaptic terminal is:

- A)** reuptake of neurotransmitter
- B)** arrival of the neurotransmitter enzyme
- C)** influx of sodium into the postsynaptic cell
- D)** influx of calcium ions into the terminal
- E)** exit of potassium from the terminal

Answer: D

The arriving action potential opens voltage-gated Ca^{2+} channels in the presynaptic terminal, and the resulting calcium influx is the direct trigger for synaptic vesicles to fuse with the membrane and release transmitter. Postsynaptic sodium entry happens afterwards as a consequence, and reuptake is the cleanup step, not the trigger.

26. Several hours after a carbohydrate-rich meal, blood glucose is falling. Which endocrine change is occurring?

- A)** insulin rises, glucagon rises
- B)** insulin falls, glucagon falls
- C)** insulin rises, glucagon falls
- D)** insulin falls, glucagon rises
- E)** neither hormone changes

Answer: D

As blood glucose falls, pancreatic beta cells secrete less insulin and alpha cells secrete more glucagon, which raises glucose via glycogenolysis and gluconeogenesis. Any combination in which insulin rises while glucose is low, or glucagon falls, would push blood glucose in the wrong direction.

27. Antivenom is produced by immunising horses, then collecting and purifying the antibodies. A snakebite victim receiving antivenom acquires:

- A)** active natural immunity
- B)** no immunity, since antibodies cannot be injected
- C)** immunity produced by the victim's own plasma cells weeks later
- D)** active artificial immunity with lasting memory
- E)** passive artificial immunity: ready-made antibodies, protection that is immediate but without immune memory

Answer: E

Antivenom is a preparation of ready-made antibodies raised in horses and injected into the victim, so protection is immediate but the victim's own immune system never produced it and keeps no memory of it: passive, artificially acquired immunity. Active immunity would take weeks for the victim's own plasma cells to respond, useless against venom, and injected antibodies obviously do confer protection.

28. A patient is blood group O, Rh-negative. Which donor red-cell type can be transfused to this patient without a risk of agglutination (within the ABO and Rh systems)?

- A)** A negative
- B)** AB negative
- C)** O negative
- D)** O positive
- E)** any type, as long as it is Rh-negative

Answer: C

A group O person already has anti-A and anti-B antibodies in the plasma, so donor red cells must carry neither A nor B; being Rh-negative adds the requirement that the cells lack the D antigen as well. Only O negative red cells satisfy both conditions at once. Any Rh-negative type is not enough, because A, B or AB cells would still be agglutinated by the recipient's anti-A or anti-B.

29. Pancreatic lipase, acting in the small intestine, digests:

- A)** vitamins into coenzymes
- B)** triglycerides (fats) into fatty acids and glycerol
- C)** starch into maltose
- D)** nucleic acids into nucleotides
- E)** proteins into amino acids

Answer: B

Lipase hydrolyses the ester bonds of dietary triglycerides, releasing fatty acids and glycerol (plus monoacylglycerols), usually acting on bile-emulsified droplets in the small intestine. Proteins are cut by proteases such as trypsin and pepsin, starch by amylase, nucleic acids by nucleases; vitamins are not digested into coenzymes.

30. Glucose that is filtered at the glomerulus is normally:

- A)** almost completely reabsorbed in the proximal convoluted tubule
- B)** excreted entirely in the urine
- C)** converted to urea in the nephron
- D)** secreted into the filtrate by the distal tubule
- E)** reabsorbed only in the collecting duct under ADH control

Answer: A

Essentially all filtered glucose is reabsorbed actively in the proximal convoluted tubule, co-transported with sodium (SGLT proteins), so normal urine is glucose-free. Glucose spills into urine only when plasma glucose exceeds the tubular transport maximum, as in uncontrolled diabetes. The collecting duct's ADH-controlled reabsorption concerns water, not glucose.

31. Which cells deposit new bone matrix during bone growth and remodeling?

- A)** osteoclasts
- B)** chondrocytes
- C)** red marrow stem cells
- D)** osteocytes in lacunae
- E)** osteoblasts

Answer: E

Osteoblasts secrete osteoid, the organic matrix that then mineralises, so they are the cells that deposit new bone during growth and remodelling. Osteoclasts do the reverse (resorb bone), osteocytes are entombed mature cells that maintain the matrix, and chondrocytes build cartilage, not bone.

32. Cells are briefly exposed to radioactively labelled thymidine. Radioactivity will be incorporated into DNA during which phase?

- A)** G0 **B)** G1 **C)** M **D)** S **E)** G2

Answer: D

Labelled thymidine is a DNA building block and is incorporated only while DNA is being replicated, which happens in S phase of interphase. G1 and G2 are gap phases with no bulk DNA synthesis, M is mitosis, and G0 cells have withdrawn from the cycle.

Section C - Chemistry (Q33-47)

33. What are the oxidation numbers of nitrogen in the ammonium ion (NH_4^+) and in the nitrate ion (NO_3^-), respectively?

- A)** -5 and $+3$
B) -3 and $+5$
C) -3 and -3
D) $+1$ and $+1$ (the average of the two nitrogens)
E) $+5$ and $+5$

Answer: B

Step 1: in NH_4^+ each H is $+1$ and the total charge is $+1$: $x + 4(+1) = +1$, so $x = -3$

Step 2: in NO_3^- each O is -2 and the total charge is -1 : $x + 3(-2) = -1$, so $x = +5$

Nitrogen is -3 in ammonium and $+5$ in nitrate; an 'average of two nitrogens' is meaningless because the ions are separate species, and nitrogen cannot show -5 .

34. Given the activity series $\text{Mg} > \text{Zn} > \text{Cu} > \text{Ag}$, which reaction occurs spontaneously at room temperature?

- A)** $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$
B) none of these reactions occurs spontaneously
C) $\text{Cu(s)} + \text{Mg}^{2+}(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{Mg(s)}$
D) $\text{Cu(s)} + \text{Zn}^{2+}(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{Zn(s)}$
E) $\text{Ag(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Ag}^+(\text{aq}) + \text{Cu(s)}$

Answer: A

A spontaneous displacement needs the metal to sit above the ion's metal in the activity series (the more reactive metal is oxidised). Zn is above Cu, so zinc loses electrons to Cu^{2+} ; zinc dissolves as Zn^{2+} while copper plates out. The other proposed reactions each require copper or silver, metals low in the series, to reduce ions of a metal above them, which does not proceed at room temperature.

35. Which bond is the most polar (largest electronegativity difference)? (EN: H 2.2, C 2.55, N 3.04, O 3.44)

- A)** C-N **B)** C-O **C)** O-H **D)** C-H **E)** N-H

Answer: C

Electronegativity differences: $\text{O-H} = 3.44 - 2.2 = 1.24$; $\text{C-O} = 3.44 - 2.55 = 0.89$; $\text{N-H} = 3.04 - 2.2 = 0.84$; $\text{C-N} = 3.04 - 2.55 = 0.49$; $\text{C-H} = 2.55 - 2.2 = 0.35$

The largest difference is in O-H, making it the most polar of the five. Atomic size plays no role; only the electronegativity difference counts.

36. Which molecule contains polar bonds but has zero net dipole moment?

- A) CHCl_3 B) CO_2 C) H_2O D) SO_2 (bent) E) NH_3

Answer: B

CO_2 contains two polar $\text{C}=\text{O}$ bonds, but the molecule is linear and symmetric, so the equal bond dipoles point in opposite directions and cancel exactly, leaving zero net dipole. Water (bent), ammonia (pyramidal), sulphur dioxide (bent) and chloroform (asymmetric substitution) all have bond dipole sums that cannot cancel.

37. 100 mL of 0.1 M HCl is mixed with 100 mL of 0.3 M NaOH. The pH of the resulting solution is:

- A) 1 B) 2 C) 7 D) 12 E) 13

Answer: E

Step 1: moles of H^+ = $0.100 \text{ L} \times 0.1 \text{ M} = 0.010 \text{ mol}$; moles of OH^- = $0.100 \text{ L} \times 0.3 \text{ M} = 0.030 \text{ mol}$

Step 2: neutralisation is 1:1, so excess OH^- = $0.030 - 0.010 = 0.020 \text{ mol}$

Step 3: total volume = 0.200 L , so $[\text{OH}^-] = 0.020 \div 0.200 = 0.1 \text{ M}$, $\text{pOH} = 1$, $\text{pH} = 14 - 1 = 13$

A pH of 7 would require exact neutralisation, but the base is in threefold excess.

38. An aqueous solution of HCl is prepared at the extraordinarily low concentration of $1.0 \times 10^{-8} \text{ M}$. At 25°C its pH is:

- A) exactly 7
B) undefined, because the concentration is below 10^{-7}
C) slightly below 7
D) slightly above 7
E) about 2

Answer: C

At 10^{-8} M the added acid is ten times weaker than the H^+ that water itself provides (10^{-7} M), so self-ionisation cannot be ignored.

Solving the charge and K_w balance ($[\text{H}^+]^2 - 10^{-8}[\text{H}^+] - 10^{-14} = 0$) gives $[\text{H}^+] \approx 1.05 \times 10^{-7} \text{ M}$, i.e. $\text{pH} \approx 6.98$: an acidic solution just below 7. The pH is perfectly well defined, and it can never cross to above 7 for an acid, so both 'exactly 7' and 'slightly above 7' contradict the added H_3O^+ .

39. 25.0 mL of H_2SO_4 solution is completely neutralised by 20.0 mL of 0.15 M NaOH ($\text{H}_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$). The molarity of the acid is:

- A) 0.03 M B) 0.06 M C) 0.12 M D) 0.15 M E) 0.30 M

Answer: B

Step 1: moles of NaOH = $0.0200 \text{ L} \times 0.15 \text{ M} = 0.0030 \text{ mol}$

Step 2: the equation needs 2 mol of NaOH per 1 mol of H_2SO_4 , so acid moles = $0.0030 \div 2 = 0.0015 \text{ mol}$.

Step 3: concentration = $0.0015 \text{ mol} \div 0.0250 \text{ L} = 0.060 \text{ M}$

Skipping the factor of 2 gives 0.12 M, the classic slip.

40. 2.50 g of a hydrate of copper(II) sulfate ($\text{CuSO}_4 \cdot n\text{H}_2\text{O}$, $M(\text{CuSO}_4) = 160 \text{ g/mol}$) is heated to constant mass, leaving 1.60 g of anhydrous CuSO_4 . The value of n is:

- A) 1 B) 2 C) 3 D) 4 E) 5**

Answer: E

Step 1: moles of $\text{CuSO}_4 = 1.60 \text{ g} \div 160 \text{ g/mol} = 0.010 \text{ mol}$

Step 2: water lost = $2.50 - 1.60 = 0.90 \text{ g} = 0.90 \div 18 = 0.050 \text{ mol}$

Step 3: $n = 0.050 \div 0.010 = 5$, so the hydrate is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

41. What is the mass of 5.6 L of oxygen gas O_2 measured at STP (molar volume 22.4 L/mol; $M(\text{O}_2) = 32 \text{ g/mol}$)?

- A) 2 g B) 4 g C) 8 g D) 16 g E) 32 g**

Answer: C

Step 1: moles of $\text{O}_2 = 5.6 \text{ L} \div 22.4 \text{ L/mol} = 0.25 \text{ mol}$

Step 2: mass = $0.25 \text{ mol} \times 32 \text{ g/mol} = 8 \text{ g}$

The value 16 g comes from doubling the molar mass as if the gas were tetra-atomic; oxygen gas is diatomic O_2 with $M = 32$.

42. The species Na^+ , Mg^{2+} , F^- , O^{2-} and Ne all have 10 electrons. Which is the smallest?

- A) Mg^{2+} B) F^- C) Na^+ D) Ne E) O^{2-}**

Answer: A

All five species share the same 10-electron shell arrangement, so size is set purely by nuclear charge: more protons pull the identical electron cloud tighter. Proton counts: O^{2-} 8, F^- 9, Ne 10, Na^+ 11, Mg^{2+} 12, so the magnesium ion is the smallest and the oxide ion the largest.

43. Which of the following can exist as cis and trans isomers?

- A) ethene
B) but-1-ene
C) propane
D) but-2-ene
E) 2-methylpropene**

Answer: D

Geometrical isomerism around $\text{C}=\text{C}$ requires each double-bond carbon to carry two different groups. In but-2-ene each alkene carbon bears H and CH_3 , so the methyls can sit on the same side or on opposite sides: cis and trans. But-1-ene and 2-methylpropene each have two identical groups on one alkene carbon (two H, or two CH_3), ethene has only hydrogens on both, and propane has no double bond at all.

44. Propan-1-ol is oxidised with acidified potassium dichromate, and the volatile organic product is distilled off as it forms. The product is:

- A) propane B) propanone C) propan-2-ol D) propanal E) propanoic acid**

Answer: D

Oxidation of a primary alcohol runs $\text{propan-1-ol} \rightarrow \text{propanal} \rightarrow \text{propanoic acid}$. Distilling the volatile product off as it forms removes the aldehyde (propanal) before the second oxidation can occur, so that is what is collected; prolonged reflux in contact with the dichromate would push on to propanoic acid. Propanone would require the secondary alcohol propan-2-ol as the starting material.

45. For the equilibrium $A \rightleftharpoons B$, a solution initially containing only A at 1.0 M reaches equilibrium with $[B] = 0.75$ M. The equilibrium constant K_c is:

- A) 0.25 B) 0.75 C) 1.33 D) 3.0 E) 4.0

Answer: D

The 1:1 stoichiometry means forming 0.75 M of B consumes 0.75 M of A, leaving $[A]_{eq} = 1.0 - 0.75 = 0.25$ M.

$$K_c = [B]/[A] = 0.75/0.25 = 3.0$$

Quoting 0.75 or 0.25 confuses an equilibrium concentration with the equilibrium constant, which is the ratio of the two.

46. For an exothermic reaction at equilibrium, increasing the temperature:

- A) decreases K_c
B) increases K_c
C) leaves K_c unchanged
D) increases K_c , which then returns to its old value
E) has an unpredictable effect without the numerical K_c

Answer: A

For an exothermic reaction heat is effectively a product, so adding heat shifts the position in the endothermic (reverse) direction and the equilibrium constant itself decreases. K_c depends only on temperature, so it is neither left unchanged nor restored to its old value; the fact that the reaction runs faster at higher temperature says nothing about where the equilibrium sits.

47. What is the approximate molarity of pure water at 25 °C (density 1.0 g/mL, $M(H_2O) = 18$ g/mol)?

- A) 0.056 M B) 1.0 M C) 18 M D) 55.6 M E) 100 M

Answer: D

One litre of pure water has mass 1000 g (density 1.0 g/mL). Moles of $H_2O = 1000 \text{ g} \div 18 \text{ g/mol} \approx 55.6$ mol, so the concentration of water in itself is about 55.6 M. The 18 M figure confuses the molar mass with the molarity of a litre.

Section D - Mathematics (Q48-54)

48. Solve $x^2 - 5x + 6 > 0$.

- A) all real x B) $-3 < x < -2$ C) $x < 2$ or $x > 3$ D) $x < 2$ only E) $2 < x < 3$

Answer: C

Factorise: $x^2 - 5x + 6 = (x - 2)(x - 3)$; the parabola opens upward with roots 2 and 3. The product is positive outside the roots (both factors positive when $x > 3$, both negative when $x < 2$), so the solution set is $x < 2$ or $x > 3$. The interval $2 < x < 3$ is where the expression is negative, the answer to the reversed inequality.

49. Solve $|2x - 1| \leq 5$.

- A) no solution B) $-2 \leq x \leq 3$ C) $x \geq -2$ D) $-3 \leq x \leq 2$ E) $x \leq 3$

Answer: B

$|2x - 1| \leq 5$ unpacks to $-5 \leq 2x - 1 \leq 5$. Adding 1 throughout: $-4 \leq 2x \leq 6$. Dividing by 2: $-2 \leq x \leq 3$. The tempting shifted interval $-3 \leq x \leq 2$ comes from forgetting to add 1 to both bounds.

50. Given $\log_{10} 2 = 0.30$, the value of $\log_{10} 5$ is:

- A) 0.50 B) 0.60 C) 0.70 D) 1.30 E) 2.30

Answer: C

$$\log 5 = \log(10 \div 2) = \log 10 - \log 2 = 1 - 0.30 = 0.70$$

Since 5 is half of 10, its log must fall just under 1; halving the log itself to 0.50 confuses the logarithm with the number.

51. Two fair dice are rolled. What is the probability of getting at least one six?

- A) $1/6$ B) $1/12$ C) $1/36$ D) $5/36$ E) $11/36$

Answer: E

Use the complement: $P(\text{no six}) = (5/6) \times (5/6) = 25/36$, so $P(\text{at least one six}) = 1 - 25/36 = 11/36$. The value $5/36$ covers only the case where one specified die shows the six and the other does not; the six can land on either die (and double sixes also count), which is why the correct value is larger.

52. A vertical pole 6 m tall casts a shadow when the sun's elevation is 45° . How long is the shadow? ($\tan 45^\circ = 1$, $\tan 30^\circ = 0.58$)

- A) 3 m B) 4.2 m C) 6 m D) 8.5 m E) 12 m

Answer: C

In the right triangle, $\tan(\text{elevation}) = \text{height} \div \text{shadow length}$, so $\text{shadow} = 6 \div \tan 45^\circ = 6 \div 1 = 6$ m. At 45° the triangle is isosceles, so the shadow exactly equals the pole's height; the $\tan 30^\circ$ figure is unused distractor data.

53. If $f(x) = (2x + 3)/5$, then $f^{-1}(x)$, the inverse function, is:

- A) $(5x - 3)/2$ B) $2x + 3/5$ C) $(2x - 3)/5$ D) $(5x + 3)/2$ E) $5/(2x + 3)$

Answer: A

Set $y = (2x + 3)/5$ and solve for x : $5y = 2x + 3$, so $2x = 5y - 3$ and $x = (5y - 3)/2$

Swapping variable names gives $f^{-1}(x) = (5x - 3)/2$

Flipping the sign to $(5x + 3)/2$, or simply restating the original formula, are the standard slips.

54. What is the vertex (turning point) of the parabola $y = x^2 - 4x + 7$?

- A) (2, 3) B) (-2, 3) C) (2, 11) D) (4, 7) E) (-2, 19)

Answer: A

Vertex x-coordinate: $x = -b \div 2a = 4 \div 2 = 2$

Then $y = 2^2 - 4(2) + 7 = 4 - 8 + 7 = 3$, so the vertex is (2, 3)

The point (2, 11) comes from computing $4 + 8 + 7$, dropping the minus on the linear term; using $x = -b = 4$ instead of $-b/2a$ produces the (4, 7) decoy.

Section E - Physics (Q55-60)

55. A 12 V battery drives a 4 Ω resistor in series with a parallel pair of two 4 Ω resistors. What power is dissipated in one of the two parallel resistors?

- A)** 2 W **B)** 4 W **C)** 8 W **D)** 16 W **E)** 24 W

Answer: B

Step 1: two 4 Ω resistors in parallel give 2 Ω , so the circuit total is 4 + 2 = 6 Ω .

Step 2: $I = 12 \text{ V} \div 6 \Omega = 2 \text{ A}$

Step 3: the parallel section takes $2 \text{ A} \times 2 \Omega = 4 \text{ V}$, so each 4 Ω resistor dissipates $P = V^2/R = 16 \div 4 = 4 \text{ W}$ (equivalently, the 2 A splits into 1 A per branch: $1^2 \times 4 = 4 \text{ W}$). The 8 W figure is the whole pair and 24 W the entire circuit; a single branch does not carry the full 2 A.

56. A stone is dropped from rest from a height of 5 m ($g = 10 \text{ m/s}^2$, air resistance negligible). Its speed just before impact is:

- A)** 5 m/s **B)** 10 m/s **C)** 20 m/s **D)** 25 m/s **E)** 50 m/s

Answer: B

From rest: $v^2 = u^2 + 2gh = 2 \times 10 \times 5 = 100$, so $v = 10 \text{ m/s}$ just before impact.

Alternatively the fall takes $t = \sqrt{(2h \div g)} = 1 \text{ s}$ and $v = gt = 10 \text{ m/s}$

The 50 m/s decoy multiplies g by h once, confusing the formula with $v^2 = 2gh$.

57. 1 kg of water and 1 kg of iron each receive 10 kJ of heat. ($c_{\text{water}} = 4200$, $c_{\text{iron}} = 450 \text{ J/(kg}\cdot\text{K)}$.) Which statement is correct?

- A)** It depends on the shape of the containers.
B) Neither changes temperature.
C) The water ends at the higher temperature.
D) Both end at the same temperature.
E) The iron ends at the higher temperature, rising about 22 K versus about 2.4 K for the water.

Answer: E

$\Delta T = Q \div (mc)$. Water: $10000 \div 4200 \approx 2.4 \text{ K}$. Iron: $10000 \div 450 \approx 22.2 \text{ K}$. With equal masses and equal heat input, the smaller specific heat capacity (iron) gives the larger temperature rise, about nine times larger; container shape does not appear in the calorimetric equation.

58. A ball falls freely from rest for 2.0 s ($g = 10 \text{ m/s}^2$). The distance it falls is:

- A)** 5 m **B)** 10 m **C)** 20 m **D)** 40 m **E)** 80 m

Answer: C

From rest, $s = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (2.0)^2 = 5 \times 4 = 20 \text{ m}$

The 10 m answer uses $\frac{1}{2}gt$ and forgets to square the time.

59. A person stands on bathroom scales inside a lift that is accelerating upward. Compared with the person's true weight:

- A)** the reading depends on the lift's velocity, not its acceleration
- B)** the scale reads zero
- C)** the scale reads less
- D)** the scale reads exactly the true weight
- E)** the scale reads more, because the floor must supply $m(g + a)$

Answer: E

Taking up as positive, Newton's second law on the person gives $N - mg = ma$, so $N = m(g + a)$, which exceeds the true weight mg : the scales read more and the person feels heavier. The reading tracks acceleration, not velocity: at constant upward speed it would read exactly the true weight.

60. A sound wave of frequency 500 Hz has a wavelength of 0.68 m in air. The speed of sound is:

- A)** 34 m/s **B)** 170 m/s **C)** 340 m/s **D)** 500 m/s **E)** 680 m/s

Answer: C

Wave speed: $v = f\lambda = 500 \text{ Hz} \times 0.68 \text{ m} = 340 \text{ m/s}$, the familiar speed of sound in air. Doubling the wavelength gives the 680 m/s decoy; halving it gives 170 m/s.